



NITTE
(Deemed to be University)

**NMAM INSTITUTE
OF TECHNOLOGY**

(An off-Campus Institution of NITTE (DEEMED TO BE UNIVERSITY), MANGALORE)

Department of MCA

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ADVANCED WEB TECHNOLOGIES LAB ASSIGNMENT 2

Cascading Style Sheets

1. Design a web page demonstrating different levels of Cascading Style Sheets (CSS) namely inline, internal, and external styles.

Show clearly how priority (precedence) is applied among these stylesheet levels.

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Priority Demo</title>

  <!-- External CSS -->
  <link rel="stylesheet" href="styles.css">

  <!-- Internal CSS -->
  <style>
    h1 {
      color: green;
      font-size: 40px;
    }

    p {
      color: purple;
    }
  </style>
</head>
```

```

<body>

  <!-- Inline CSS ->
  <h1 style="color: red; font-size: 50px;">
    CSS Priority Example
  </h1>

  <p>This paragraph is styled using internal CSS.</p>

</body>
</html>

```

styles.css

```

h1 {
  color: blue;
  font-size: 30px;
}

```

2. Develop an HTML page to demonstrate the use of different CSS selectors such as element selector, class selector, ID selector, group selector, and universal selector.

index.html

```

<!DOCTYPE html>
<html>
<head>
  <title>CSS Selectors Demo</title>

  <style>
    /* Universal Selector */
    * {
      font-family: Arial, sans-serif;
      background-color: #f4f4f4;
    }

    /* Element Selector */
    h1 {
      color: darkblue;
    }
  </style>

```

```
/* Class Selector */
.highlight {
  color: green;
  font-weight: bold;
}

/* ID Selector */
#unique {
  color: red;
  font-size: 20px;
}

/* Group Selector */
h2, p {
  color: purple;
}
</style>

</head>
<body>

  <h1>CSS Selectors Demonstration</h1>

  <h2>Group Selector Example</h2>
  <p>This paragraph is styled using the group selector.</p>

  <p class="highlight">
    This paragraph uses a class selector.
  </p>

  <p id="unique">
    This paragraph uses an ID selector.
  </p>

  <p>
    This is a normal paragraph (only group + universal apply).
  </p>

</body>
</html>
```

3. Create a web page illustrating the application of font and text-related CSS properties, including font-family, font-size, font-style, font-weight, color, text-align, and text-decoration.

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Font and Text Properties Demo</title>

  <style>
    body {
      font-family: Arial, sans-serif; /* font-family */
      background-color: #f2f2f2;
    }

    h1 {
      text-align: center;          /* text-align */
      color: darkblue;            /* color */
      font-size: 36px;            /* font-size */
    }

    .italic-text {
      font-style: italic;          /* font-style */
      font-size: 18px;
    }

    .bold-text {
      font-weight: bold;           /* font-weight */
      color: green;
    }

    .decorated-text {
      text-decoration: underline;  /* text-decoration */
      color: red;
      font-size: 20px;
    }

    .mixed-style {
      font-family: "Courier New", monospace;
      font-size: 22px;
      font-style: italic;
      font-weight: bold;
      color: purple;
    }
  </style>
</head>
<body>
  <h1>Font and Text Properties Demo</h1>
  <p>This page demonstrates various font and text CSS properties. The background is light gray (#f2f2f2). The main heading is centered, dark blue, and 36px. The first paragraph is italicized and 18px. The second paragraph is bold and green. The third paragraph has a red underline, is 20px, and red. The fourth paragraph is in a monospace font, italicized, bold, 22px, and purple.
  </p>
</body>
</html>
```

```

        text-align: right;
        text-decoration: line-through;
    }
</style>
</head>

<body>

    <h1>CSS Font and Text Properties</h1>

    <p class="italic-text">
        This paragraph demonstrates the font-style property (italic).
    </p>

    <p class="bold-text">
        This paragraph demonstrates the font-weight property (bold).
    </p>

    <p class="decorated-text">
        This paragraph demonstrates text-decoration (underline).
    </p>

    <p class="mixed-style">
        This paragraph combines multiple font and text properties.
    </p>

</body>
</html>

```

4. Design an HTML page to demonstrate list-related CSS properties such as list-style-type, list-style-position, list-style-image, and list-style.

index.html

```

<!DOCTYPE html>
<html>
<head>
    <title>CSS List Properties Demo</title>

    <style>
        body {

```

```
    font-family: Arial, sans-serif;
    background-color: #f2f2f2;
    padding: 20px;
}

h1 {
    text-align: center;
}

section {
    background: white;
    padding: 15px;
    margin: 20px 0;
    border-radius: 8px;
}

/* 1. list-style-type */
.type-example ul {
    list-style-type: square;
}

.type-example ol {
    list-style-type: upper-roman;
}

/* 2. list-style-position */
.position-inside {
    list-style-position: inside;
}

.position-outside {
    list-style-position: outside;
}

/* 3. list-style-image */
.image-example {
    list-style-image: url("bullet.jpg");
}

/* 4. list-style (shorthand) */
.shorthand-example {
    list-style: circle inside;
}
```

```
</style>
</head>

<body>

  <h1>CSS List Properties Demonstration</h1>

  <section class="type-example">
    <h2>1. list-style-type</h2>
    <p>Unordered List (square):</p>
    <ul>
      <li>HTML</li>
      <li>CSS</li>
      <li>JavaScript</li>
    </ul>

    <p>Ordered List (upper-roman):</p>
    <ol>
      <li>Frontend</li>
      <li>Backend</li>
      <li>Database</li>
    </ol>
  </section>

  <section>
    <h2>2. list-style-position</h2>

    <p><strong>Inside:</strong></p>
    <ul class="position-inside">
      <li>This text wraps inside the bullet alignment.</li>
      <li>Notice how the text aligns.</li>
    </ul>

    <p><strong>Outside (default):</strong></p>
    <ul class="position-outside">
      <li>This text wraps outside the bullet alignment.</li>
      <li>This is the default behavior.</li>
    </ul>
  </section>

  <section>
    <h2>3. list-style-image</h2>
```

```

    <ul class="image-example">
      <li>Custom Bullet 1</li>
      <li>Custom Bullet 2</li>
      <li>Custom Bullet 3</li>
    </ul>
  </section>

  <section>
    <h2>4. list-style (Shorthand)</h2>
    <ul class="shorthand-example">
      <li>Circle + Inside</li>
      <li>Defined using shorthand</li>
      <li>Cleaner CSS</li>
    </ul>
  </section>

</body>
</html>

```

5. Create a web page to demonstrate the CSS Box Model. Show the effect of margin, border, padding, width, and height using suitable elements.

```

index.html
<!DOCTYPE html>
<html>
<head>
  <title>CSS Box Model Demonstration</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
    }

    h1 {
      text-align: center;
    }

    .box {
      width: 300px;          /* Width of content */
      height: 150px;        /* Height of content */
    }
  </style>

```



```

padding: 20px;          /* Space inside border */
border: 5px solid blue; /* Border around padding */

margin: 40px auto;      /* Space outside border */

background-color: lightyellow;
text-align: center;
}

.label {
background-color: lightgreen;
padding: 5px;
margin: 10px;
}
</style>
</head>

<body>

<h1>CSS Box Model Demonstration</h1>

<div class="box">
  <div class="label">Content Area</div>
  This box demonstrates width, height, padding, border, and margin.
</div>

</body>
</html>

```

6. Develop a web page illustrating the use of <div> and tags to apply block-level and inline-level styling using CSS.

index.html

```

<!DOCTYPE html>
<html>
<head>
  <title>Div and Span Demonstration</title>

```

```

<style>
  body {
    font-family: Arial, sans-serif;
    background-color: #f4f4f4;
  }

  h1 {
    text-align: center;
  }

  /* Block-level styling using div */
  .block-box {
    background-color: lightblue;
    border: 2px solid blue;
    padding: 15px;
    margin: 15px 0;
  }

  /* Inline-level styling using span */
  .highlight {
    color: red;
    font-weight: bold;
    background-color: yellow;
    padding: 2px 5px;
  }
</style>
</head>

<body>

  <h1>Block-level and Inline-level Styling</h1>

  <!-- Block-level example -->
  <div class="block-box">
    This is a DIV element. It is a block-level element,
    so it takes up the full width available and starts on a new line.
  </div>

  <div class="block-box">
    This is another DIV element. Notice it appears on a new line automatically.
  </div>

  <!-- Inline-level example -->

```

```
<p>
  This is a paragraph with a
  <span class="highlight">SPAN element</span>
  inside it. The span only styles part of the text and does not start on a new line.
</p>

</body>
</html>
```

7. Design an HTML page to demonstrate background-related CSS properties such as background-color, background-image, background-repeat, and background-position.

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Background CSS Properties Demo</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #e6f2ff; /* background-color */
    }

    h1 {
      text-align: center;
    }

    .box1 {
      width: 300px;
      height: 150px;
      margin: 20px auto;
      background-color: lightgreen; /* background-color */
      text-align: center;
      line-height: 150px;
      border: 2px solid black;
    }

    .box2 {
```

```

        width: 300px;
        height: 150px;
        margin: 20px auto;
        background-image: url("bg2.jpeg");
        background-repeat: repeat;    /* background-repeat */
        border: 2px solid black;
        text-align: center;
        line-height: 150px;
    }

    .box3 {
        width: 300px;
        height: 150px;
        margin: 20px auto;
        background-image: url("bg.jpeg");
        background-repeat: no-repeat; /* background-repeat */
        background-position: center; /* background-position */
        border: 2px solid black;
        text-align: center;
        line-height: 150px;
    }
</style>
</head>

<body>

    <h1>Background Properties Demonstration</h1>

    <div class="box1">
        background-color
    </div>

    <div class="box2">
        background-image + repeat
    </div>

    <div class="box3">
        no-repeat + center position
    </div>

</body>
</html>

```

8. Create a web page demonstrating CSS border properties, including border style, width, color, and radius.

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Border Properties Demo</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f4f4;
      text-align: center;
    }

    h1 {
      margin-bottom: 30px;
    }

    .box {
      width: 250px;
      padding: 20px;
      margin: 20px auto;
      background-color: white;
    }

    /* Border style */
    .style-demo {
      border-style: dashed;
      border-width: 4px;
      border-color: blue;
    }

    /* Border width */
    .width-demo {
      border-style: solid;
      border-width: 8px;
      border-color: green;
    }
```

```

/* Border color */
.color-demo {
    border-style: solid;
    border-width: 5px;
    border-color: red;
}

/* Border radius */
.radius-demo {
    border: 5px solid purple;
    border-radius: 25px;
}
</style>
</head>

<body>

    <h1>CSS Border Properties Demonstration</h1>

    <div class="box style-demo">
        Border Style (dashed)
    </div>

    <div class="box width-demo">
        Border Width (8px)
    </div>

    <div class="box color-demo">
        Border Color (red)
    </div>

    <div class="box radius-demo">
        Border Radius (rounded corners)
    </div>

</body>
</html>

```

9. Create a navigation menu using HTML and CSS, applying hover effects.

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Navigation Menu with Hover Effect</title>

  <style>
    body {
      margin: 0;
      font-family: Arial, sans-serif;
    }

    /* Navigation Bar */
    nav {
      background-color: #222;
    }

    /* Remove default list styling */
    ul {
      list-style: none;
      margin: 0;
      padding: 0;
      display: flex; /* Makes menu horizontal */
    }

    /* Menu items */
    li {
      margin: 0;
    }

    /* Links */
    a {
      display: block;
      padding: 15px 25px;
      color: white;
      text-decoration: none;
      transition: background-color 0.3s, color 0.3s;
    }

    /* Hover effect */
    a:hover {
      background-color: #00bcd4;
      color: black;
    }
  </style>
```

</head>

<body>

<nav>

Home

About

Services

Contact

</nav>

</body>

</html>